Ibasho 居場所の向上

Status, explainer and walkthrough

Progress

Phase	State	Notes
1 · Research and direction	DONE	Eight research briefs, plan v2 (VISION, SPEC, ARCHITECTURE, DESIGN_SYSTEM “Andon”, BUILD_PLAN), three red-team reviews, rulings in `docs/review/DECISIONS.md`.
2 · Scaffold and walking skeleton	ACTIVE	G0 monorepo and contracts, then M-0: one thin real slice through every layer.
3 · Parallel build	TODO	Engine, proofs, AI, UI kit, 3D, app shell, assets, landing.
4 · Integration and QA	TODO	Tests, e2e, red-team, polish.

What Ibasho is

Ibasho is a digital twin of a home that you can edit. Change the house and the model recomputes it, using a documented reduced-order heat balance [1], [2]. It searches the retrofit packages you can actually build and shows the best few. Lean 4 [3] proves the bookkeeping. An AI narrator explains results but cannot change a number. The visual language takes cues from architectural storytelling [4].

Features

Feature	State
Planned. Filled in as work packages land.	TODO
Agent link: MCP server and Claude Code plugin, so an AI agent can explore a home headless or live in the app. One-command install.	TODO
G0 scaffold: a 28-package monorepo with enforced layers and ownership. The kernel holds every contract plus exact fixed-point money, exact fractions for the score, SHA-256 hashing [5], a seeded generator [6] and Latin hypercube sampling [7], all tested. Both apps boot in Chromium and Firefox.	DONE
Andon UI kit, first slice: day and night colour tokens checked for contrast and colour blindness [8], [9], three self-hosted font families, seal kanji baked from Yuji Syuku [10], and the number chip, check seal and proved claim seal.	DONE
Physics, walking-skeleton slice: GH1, a 1905 London terrace, runs through a steady monthly heat balance [2], [11]. It models fabric and ventilation losses, solar and internal gains, hot water, boiler efficiency and a fixed heat pump COP, all on London climate data [12]. Three measures (loft, draught-proofing, heat pump) patch the twin, and a package is always recomputed whole. Maths is deterministic, ported from fdlibm [13]. Energy is rounded to whole watt hours exactly once. The fast path for heating changes matches a full rerun bit for bit.	ACTIVE
Hero assets: four Poly Haven skies [14], four material sets and 38 pieces of Kenney [15] and Quaternius [16] furniture for the GH1 terrace, all CC0. Skies ship as compact gain maps [17], models in metres with meshopt. Every file is listed with its licence and checksum, and the credits page is generated from that list. 5.8 MB of the 8 MB budget.	DONE
Rules, walking-skeleton slice: rule conditions are data, evaluated in three-valued logic [18], so an unknown fact is never read as “no”. A package blocked only by something we do not know goes to “possible after a check”, never to the front. Two rules run on GH1: the budget, which does nothing until a budget is set, and a heat pump on a possibly shared electricity supply,	ACTIVE

which waits for the network operator to approve. A randomised test checks that an unknown block is never feasible.

Search and finance, walking-skeleton slice: GH1′s 8 packages (loft, draught-proofing, heat pump) are enumerated in full and costed. Bills are exact at the October 2026 price cap [19], with standing charges. Carbon is counted per fuel and month with SAP 10.2 factors [2]. Grants include the Boiler Upgrade Scheme [20]. Upfront cost and NPV at the Green Book 3.5% [21] are in whole micro-pence with named rounding. The best options form a front, every other option has a named witness that beats it, and the three picks follow the stated tie-breaks. The search runs in a worker, reports progress and stops on cancel. It emits the certificate input for the Lean checker, and the same request always gives the same hash. A property-value table [22] is ready but not yet shown. ACTIVE

Lean checker and verify, walking-skeleton slice: a checker in Lean 4 [3] re-derives every saving, carbon figure, upfront cost, NPV, rule status, front, witness and pick of a certificate from the physics and pack data, and a machine-checked theorem says that passing every check means the claims hold. The constants the app uses, the certificate types and the claim seals with each theorem's statement are generated from Lean, with a SHA-256 spec hash [5], and CI fails if they drift. A TypeScript mirror gives the same verdicts. On GH1 both checkers pass the real certificate and both catch three planted errors: a flipped saving, a rejected measure on the front, and a heat pump with an unknown supply marked feasible. DONE

Words and landing page: plain guidance for all 16 home measures (how many days of work, which rooms, whether you can stay in, who to call, which accreditation to look for, and three questions to ask before you sign) [23], [24], [25], [26], a glossary of about 50 terms, and a voice checker that rejects "guaranteed" and "save up to". The landing page paints as static HTML in five chapters, and a chapter appears only once the feature it describes exists. DONE

Lean spec, first slice: the bookkeeping rules are written once in Lean 4 [3] and proved for every input. Exact rounding to the nearest even unit, discounting, per fuel and per month savings and carbon, the score (weights add to one, always 0 to 100, never lower for a better home), three-valued rule logic [18] (an unknown is never read as a pass), the heat pump noise test of MCS 020 a) [27] decided only when rounding cannot change it, the Pareto front (nothing better is hidden), the three picks, and memory (a measure you ruled out never comes back). The build fails if any proof is missing or uses an extra axiom. ACTIVE

Agent link, first slice: an AI agent such as Claude Code can explore a home through 19 MCP tools [28], headless or live in the open app. It reads numbers only as evidence with their ids, changes a home only with typed operations, and records memory only in the person's own words. A raw number in its narration is refused, following proof-carrying numbers [29]. The live link is local-only, with a per-launch token and a one-time pairing code. It installs into Claude Code as a plugin [30], or as a skill, subagent and hook, and uninstalls without a trace. Headless numbers now come from the real engine: a search through the agent link gives the same result, hash for hash, as the app's engine on the reference home. ACTIVE

Grounded narration and memory: the story of a home is told from a template that is the same byte for byte on every run, with numbers only as evidence chips. A local model on your computer (Ollama [31]) may polish the words, but every sentence is checked by eight grounding rules and falls back to the template if it writes a number, misreads a sign, recommends something you ruled out or copies a note. Notes from previous owners are read by a quarantine that keeps only typed facts; text sent to a model is datamarked [32] and logged in an outbound ledger. 60 injection notes plant 0 numbers and 0 memories. The memory of what you told us replays 500 random sessions without ever offering a ruled-out measure, and forgetting deletes for real after 30 days. An optional companion [33] on your machine adds the Lean checker and an Ollama bridge [34]. ACTIVE

The app, walking-skeleton slice: the sample home opens at `/app/home/gh1`. The engine works in the background while the house is drawn in pencil; then it is inked, and three futures ACTIVE

appear: least upfront, best return and lowest running carbon. Every figure can be tapped to see how we know it. A second checker redoes the sums and the page shows a Checked seal only when it agrees, next to the proved promise that anything you rule out never comes back. Results are kept on your device only.

Credits: one acknowledgements file names every library, ported algorithm, data source, asset and idea Ibasho builds on, with its licence. Every agent adds to it. DONE

Walkthrough

Written once the app runs end to end.

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